

Fantasy Premier League : An optimization approach to point prediction and team selection

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Abstract: The Fantasy Premier League (FPL), the official fantasy football platform of the English Premier League, engages over 11 million participants in active seasonal competition. This study leverages diverse data analytics approaches to optimize FPL team selection. Initially, we analyse the influence of player’s skills and current form on predicting their future fantasy points performances. Build and compare three machine learning models to predict points of each players. For doing that ridge regression model, random forest regression model, and xg boost regression model have been utilized. An innovative optimization algorithm is tailored to accommodate the distinct constraints of various fantasy competitions, aiming to identify ideal team configuration for maximizing prospective points. An alternative approach employs the sharpe ratio to enhance risk adjusted returns. Empirical validation of research findings is conducted using the latest FPL season, offering a benchmark for comparison against actual player choices. This study sheds light on the complexities of optimal team curation in the context of fantasy premier league, encompassing predictive analytics and strategic optimization.

Keywords and phrases: Fantasy Premier League, Football, Ridge Regression, Random Forest, XG Boost, Optimization, Sharpe Ratio.

1. Introduction

The fantasy premier league (FPL), with over 11 million active players per season, epitomizes the surging popularity of fantasy sports, particularly in the realm of english premier league football. This dynamic virtual competition requires participants to curate real-world player selections, earning points based on actual match performances. The process of engaging with the FPL entails several structured steps for players. In order to participate, individuals are required to adhere to specific guidelines and rules established by the official FPL platform. The following summary elucidates the sequential process, delineating the salient aspects of FPL participation, encompassing player selection, composition of the playing eleven, and designation of captains and bench selection, and elucidating the scoring criteria. Engaging in the Fantasy Premier League involves several steps. First, players create a team within a budget of 100 million, picking 15 players, including goalkeepers, defenders, midfielders, and forwards, with a limit of three players from each team. Next, they choose their starting eleven players for each game week, making sure to have at least one goalkeeper, three defenders, and one forward. They also decide on substitutes in case some players don’t play. A crucial decision is picking a captain and a vice-captain. The captain earns double points, and if the captain doesn’t play, the vice-captain takes over. Points are earned based on players’ real match performances, like goals, assists, and clean sheets, with bonus points for exceptional displays. Negative actions like own goals and cards can result in losing points as mentioned in Table 1: FPL Points Scoring System

Action	Points
For playing upto 60 minutes	1
For playing 60 minutes or more (excluding stoppage time)	2
For each goal scored by a goalkeeper or defender	6
For each goal scored by a midfielder	5
For each goal scored by a forward	4
For each goal assist	3
For a clean sheet by a goalkeeper or defender	4
For a clean sheet by midfielder	1
For every three shots saves by goalkeeper	1
For each penalty save	5
For each penalty miss	-2
Bonus points for best players in a match	1-3
For every 2 goals conceded by a goalkeeper or defender	-1
For each red card	-3
For each yellow card	-1
For each own goal	-2

Players can make changes to their teams using transfers. They get unlimited transfers before the first gameweek and one free transfer each following gameweek. If more transfers are made, points are deducted. Transfers need to be done before the gameweek’s deadline, usually 90 minutes before the first match. In summary, FPL participation involves building a team within a budget, selecting a starting eleven and substitutes, designating a captain, and earning points based on players’ real performances. Transfers can be made with limits to keep in mind. Understanding these steps helps players engage effectively in the FPL competition.

2. Literature review

The FPL’s escalating recognition has intensified competition, necessitating a data analytics approach for successful team selection. Utilizing historical player performance, upcoming match complexities, and sentiment analysis from diverse sources, participants refine predictions for player’s future outcome [O’Brien, Gleeson and O’Sullivan \(2021\)](#). This data-drive methodology has become essential to maintaining a competitive edge.

Investigations into FPL dynamics reveal correlations in manager’s rank across multiple seasons, emphasizing the significance of long-term planning and and effective decision-making for success. Notably, shared decision-making strategies among successful managers, leading to ”template” teams, suggest an element of collective intelligence in the game.

Several studies have focused on leveraging machine learning models to predict player performance in Fantasy Premier League (FPL).

[Pokharel et al. \(2022\)](#) proposed an XGBoost regression model to predict player performance in Fantasy Premier League (FPL) using a publicly available dataset of FPL statistics from the 2016-17 to 2021-22 seasons. Initially, the researchers calculated each player’s return on investment (ROI) and developed four XGBoost regression models, one for each playing position: goalkeepers, defenders, midfielders, and forwards, to predict individual performance. A primary limitation of the study is its reliance on players’ total points to calculate ROI, which may overlook consistency in performance. Players with high total points may still have inconsistent performances, making them a potentially risky choice for FPL teams. Furthermore, the study did not account for player chemistry or the influence of teammates’ performance, which could impact individual outcomes. [Ramdas \(2022\)](#) explored the application on convolutional neural networks (CNNs) to predict the performance of football players in the fantasy premier league. The analysis identified optimal CNN configurations for different football positions and compared CNN performance with MLP and LSTM RNN models, showing that CNN outperformed both in terms of mean squared error. It also emphasized the effectiveness of time-series representations for improving model performance. A drawback in this paper can be identified in this paper is that it discussed various data pre-processing steps, including handling duplicate

data and missing values. However, removing players with missing gameweeks can reduce the dataset size, potentially affecting the model's ability to generalize.

Bangdiwala et al. (2022) employed three machine learning models: linear regression, decision tree, and random forest. It considers various features such as fixture difficulty, team form, player creativity, and threat levels to make these predictions. Results indicated that the random forest regression model outperformed the other two models, with lowest RMSE and MAE for both training and test datasets. One of the major drawbacks of the proposed solution in the paper is that the results are provided for overfitted and hence may not be reliable. Moreover, they have not considered to predict the points for new players in the league at least till they gain significant data on them.

Rajesh et al. (2022) utilized an open-source repository for historical FPL data from the 2016-17 season. They developed random forest and gradient boosting regressor models on the position-wise data (i.e., GK, MID, FWD, DEF) to calculate expected points for players. Afterward, they applied integer linear programming to determine the best team for a particular week.

Haugen (2017) introduced a Nash equilibrium-based approach for team selection to assist football managers in maximizing win probability and, in turn, profit. However, a key limitation of this approach is that it selects players from the entire pool without considering the individual risk factors associated with each player. Bangdiwala et al. (2022) and colleagues built linear regression, decision tree, and random forest models and evaluated these based on RMSE (Root Mean Squared Error) and found that linear regression had the lowest RMSE, suggesting it performed better than the others in their comparison. However, there might be a concern with this approach, particularly regarding new players who lack historical data.

Gupta (2019) developed both recurrent neural network (RNN) and auto-regressive integrated moving average (ARIMA) models to predict FPL player points. These models were based on datasets from seasons 2013-14 to 2015-16, with predictions extending to the 2016-17 season.

Lutz (2015) developed two models: support vector machine learning regression and neural networks for predicting the fantasy football scores of NFL players, using dataset spanning 2009 to 2014.

Matthews, Ramchurn and Chalkiadakis (2012) proposed a belief-state Markov Decision Process (MDP) for sequentially optimal team selection, which captures uncertainty in players' contributions. Moreover given the complexity of the domain and the large number of choices, the authors introduced a Bayesian Q-Learning (BQL) algorithm, which is computationally feasible and can effectively handle uncertainty in this domain. Consequently, BQL and other uncertainty-agnostic approaches were utilized on the FPL dataset for the 2010/11 season, and the results show that BQL outperforms other approaches in terms of mean final score.

A mixed-integer programming model was proposed by Beliën, Goossens and Van Reeth (2017) based on historical data to predict fantasy points for a wide variety of fantasy sports games. The model was applied to real-life fantasy cycling games, using data from 2011 to 2015.

The FPL performance predictions mentioned above predominantly rely on historical statistical data. This data encompassed player's past performances, goals, assists, and clean sheets. However, this approach offered insights into player's historical performance trends, it lacked real-time external factors that could significantly impact a player's future performance.

To address this limitation, multi-stream data analysis emerged, incorporating various external elements such as injuries, managerial decisions, fixture difficulty ratings, and insights from the betting market.

Bonello et al. (2019) stated in his paper that the accuracy of the forecasts generated by statistical predictors is greatly increased with the incorporation of betting data, news articles, and blogs. But when the both pieces of information are considered at once, the predictions become far more accurate than they would have been otherwise. While prediction accuracy slightly declines (from 91% to 89%), the average number of points scored per game week increases noticeably (from 54 to 64 points per game week). Out of 6.5 million players, this is considerable boost in points, moving the player's position on the FPL leader board from rank 800,000 (top 91%) to rank 30,000 (top 0.5%). While analysing the addition of social media data via Twitter, it encountered numerous difficulties that resulted in unreliable predictions that negatively impacted analysis's overall predictive power. FPL-specific social media could have a very positive effect on our model by developing a custom corpus based on prior years of FPL data. This is a result of the profusion of talks and many viewpoints seen on such websites.

The common needs from this literature review that we could find that there is need for a more robust but less computationally heavy approach to deliver dependable results and also, need to solve for the players who appear in the league for the first time without any historical data.

3. Data and exploratory analysis

The following two publicly available datasets, maintained by Vaastav Anand of FPL, from the seasons 2020-21, 2021-22, and 2022-23, have been used for both training and testing. Additionally, these datasets have been employed for team selection using optimization techniques.

(a) **FPL Player Statistics**

This data set constitutes with 76317 rows and 37 columns, contains information about FPL point statistics and player performance statistics that has been created from the original gameweek by gameweek data, Where each season runs over 38 weeks. Some variables in the data set are as follows:

- season: which season the match took place(e.g.,2020-21,2021-22 or 2022-23)
- name: name of the player
- position: four different positions GK,DEF,MID,FWD correspond to goal keeper, defender, midfielder, and forward.
- team: team name for which the player plays.
- xP: expected points scored by the player for the Gameweek provided by Opta.
- assists: total number of assists (i.e., the last pass before a goal is scored) by the player in the match.
- bonus: bonus points scored by the player in the match.
- bps: bonus points score based on the performance of the player in the match.
- clean_sheets: binary, indicating whether the team scored no goal during the time the player played in the match (if minutes is more than 60).
- element: unique player id for the player in the season.
- expected_assists, expected_goals, expected_goals_conceded, expected_goal_involvements: expected assists, goals scored, goals conceded, goals involvements(i.e. sum of expected assists and expected goals) for the player during the match.
- fixture: unique fixture id during the match.
- goals_conceded: number of goals conceded when the player was in the field. The clean_sheets column is 1 if and only if 'goals_conceded' is 0 and 'minutes' is above 60.
- goals_scored: number of goals scored by the player during the match.
- influence, creativity, threat, ict_index: quantifiable measures reflecting the influence, creativity, threat, and summary index of the player in the match. influence refers to the degree to which that player has made an impact on a match or matches over the season. creativity refers to a player's ability to produce goal-scoring opportunities for others. threat refers to a player's potential to score goals; it therefore gauges those individuals most likely to score goals.
- kickoff_time: kickoff date and time for the match.
- minutes: total number of minutes played by the player during the match.
- opponent_team: opponent team id of the team playing against the player's team during the match.
- own_goals: number of own goals scored by the player.
- penalties_missed, penalties_saved: numbers of penalties missed and saved (only valid for goalkeepers) by the player.
- red_cards, yellow_cards: number of red and yellow cards seen by the player.
- round: gameweek of the match, can be anything between 1 and 38 since there are 38 matches within a season.
- saves: number of saves made by player (only valid for goalkeepers).
- selected: number of fantasy players who selected the player in their teams.
- starts: total number of matches started by the player during the season.
- team_a_score, team_h_score: scoreline (for away and home teams) of the match.

- `total_points`: number of FPL points scored by the player based on the scoring criteria mentioned in Table 1: FPL Points Scoring System.
- `transfers_in`, `transfers_out`, `transfers_balance`: numbers of transfers made by players into their team, out of their team and net balance.
- `value`: price of the player before the match started in terms of
- `was_home`: binary, indicating whether the match was at home or away.

(b) **Team Statistics**

Another dataset named as team statistics is the summary file of each of the 25 teams that played in the Premier League in the three mentioned seasons. The data set consists of 25 files, each with a total of 25 columns and maximum of 114 rows representing the 38 matches played in each of the three seasons. Rows are lesser in case the team was relegated or promoted during the mentioned time period.

- `h_a`: binary, indicating whether the match was at home or away.
- `xG`: expected goals metric, it is a statistical measure of the quality of chances created and conceded.
- `xGA`: expected goals against.
- `npxG`, `npxGA`, `npxGD`:
- `ppda`: passes allowed per defensive action in the opposition half (power of the pressure).
- `ppda_allowed`: opponent passes allowed per defensive action in the opposition half (power of opponent's pressure).
- `deep`: passes completed within an estimated 20 yards of goal (crosses excluded).
- `deep_allowed`: opponent passes completed within an estimated 20 yards of goal (crosses excluded).
- `scored`: number of goals scored by the team.
- `missed`: number of goal chances missed by the team ending with a shot at opponent goal.
- `xpts,pts`: expected and actual points between 0 and 3, indicating Premier League table points based on expected result, where win = 3 points, draw = 1 point, and loss = 0 point.
- `result`: categorical, indicating either win(w), draw(d), or loss(l)
- `wins`, `draws`, `loses`: binary, indicating if the match ended in win, draw or loss for the team.
- `date`: date and time of kick-off of the match.

4. Methodology

Some high-level steps involved in this paper as follows:

- (a) Pre-process data to ensure sufficient information for regression analysis.
- (b) Analyse the correlation between player skill represented by player price and player performance in the game.
- (c) Investigate the impact of a player's and team's current form on their future fantasy performance.
- (d) Develop a predictive model that utilizes the important features to forecast fantasy points for upcoming matches.
 - i. Regression modelling for all players.
 - ii. Clustering to estimate points for new players with no FPL history.

4.1. Data pre-processing and visualization

Since most players are unable to play in every game due to benching or injury, the distribution of the outcome variable, or the points achieved by individual players, is skewed to the right with a median zero. This indicates that at least half of the players score zero points or fewer.

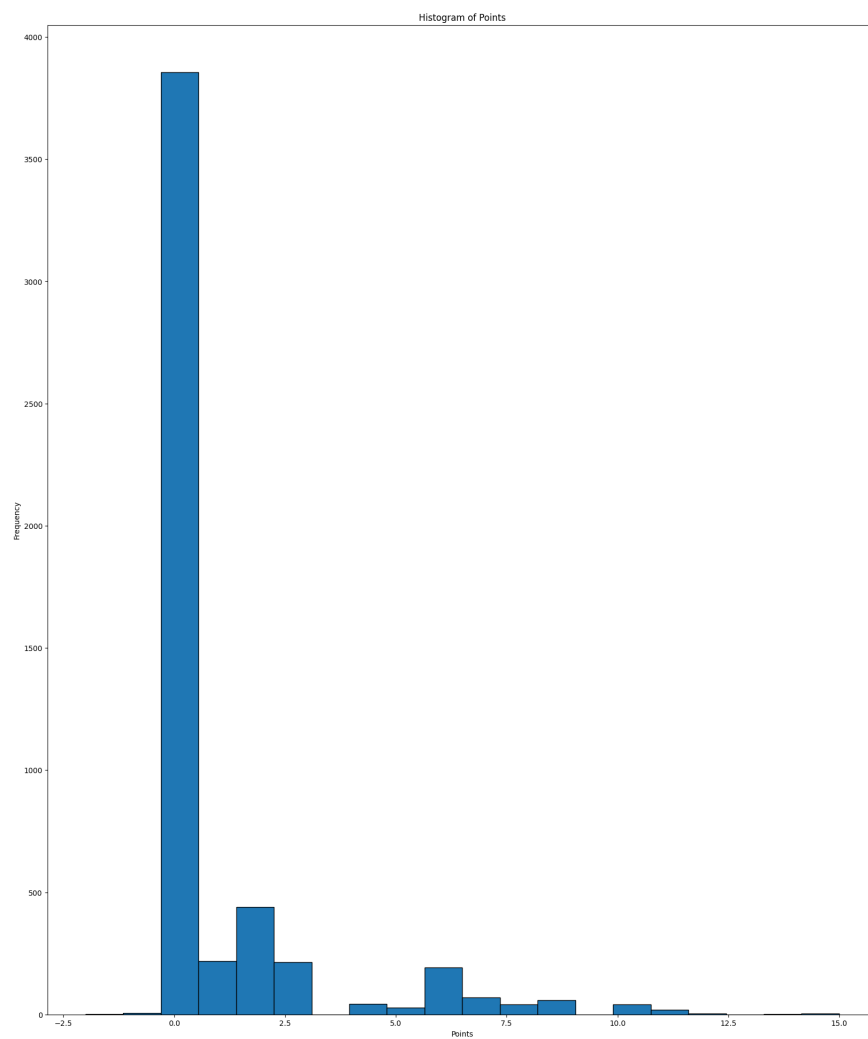


Fig 1: Points Histogram

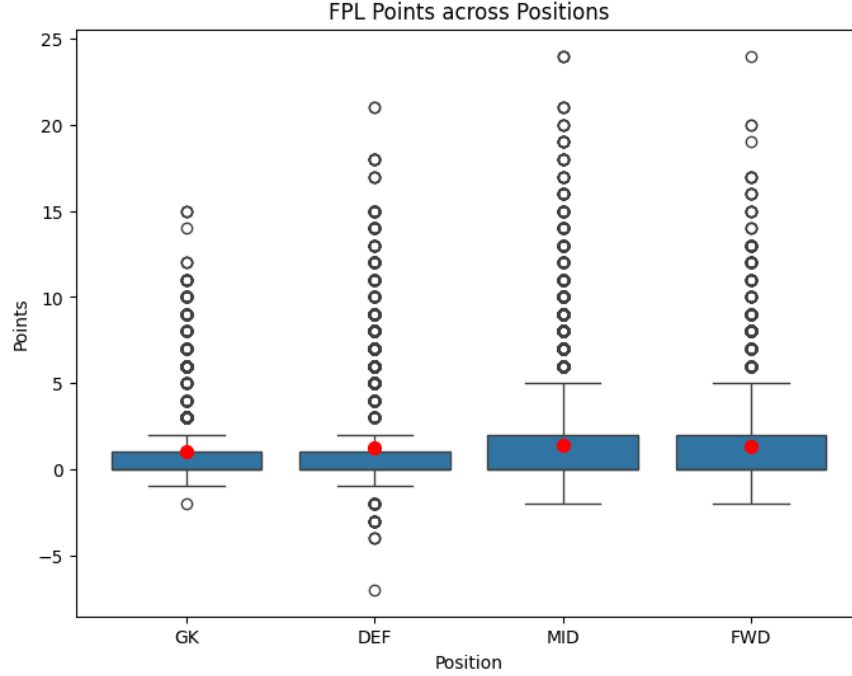


Fig 2: Points Distribution with Player Position

The relationship between the points and position shows a lower mean for goalkeepers with a lower standard deviation. Since goalkeepers have limited opportunities to score goals or make assists, their point-scoring potential is inherently lower, and their performance are less volatile because they don't experience the dramatic point swing seen in outfield players who might score multiple goals in a single match, which leads to a lower standard deviation. For positions such as forwards and midfielders have a much higher upside in points earned while defenders have a higher downside risk due to higher chances of fouls and bookings.

The player value in FPL is a representation of the player's skill and popularity among all FPL users. Over the years, the same players have been priced higher or lower based on their previous season's performance. Also, prices are dynamic changing basis supply and demand in the game. From the distribution of points across different player values, it can be inferred that, on average a higher-valued player guarantees better points however, the risk is also high given the higher standard deviation. Moreover, in few instances, the cheaper players may also perform equally well and hence, rises the need to use a value for money concept during team selection. In the given data set, only features such as position, home/away, team, opponent team and their respective difficulty rating, selected, transfer in/out and value are known before the match. Hence, only these features can be used to predict the points for the future gameweek.

However, using the concept of player and team form, it is possible that previous gameweek's performance could help determine this gameweek's performance. Hence it is important to test autocorrelation as well. To do so, we have introduced new columns into the data set through data manipulation. For each player, match performance data such as `ict.index`, goals scored, etc from the previous four gameweeks has been added to every row except the first four gameweeks in season 2020-21, as no information is available before this time. Positions and teams have been replaced with nominal values i.e. for positions, GK, DEF, MID, and FWD have been replaced by 1, 2, 3, 4 respectively. For codifying the teams, the Premier League points table for seasons 2020-21 and 2021-22 has been summarized and ranked in ascending order i.e Manchester City is ranked 25 and Bournemouth yet to play in 2022-23 is ranked 1. The total numbers of features with modification is 154.

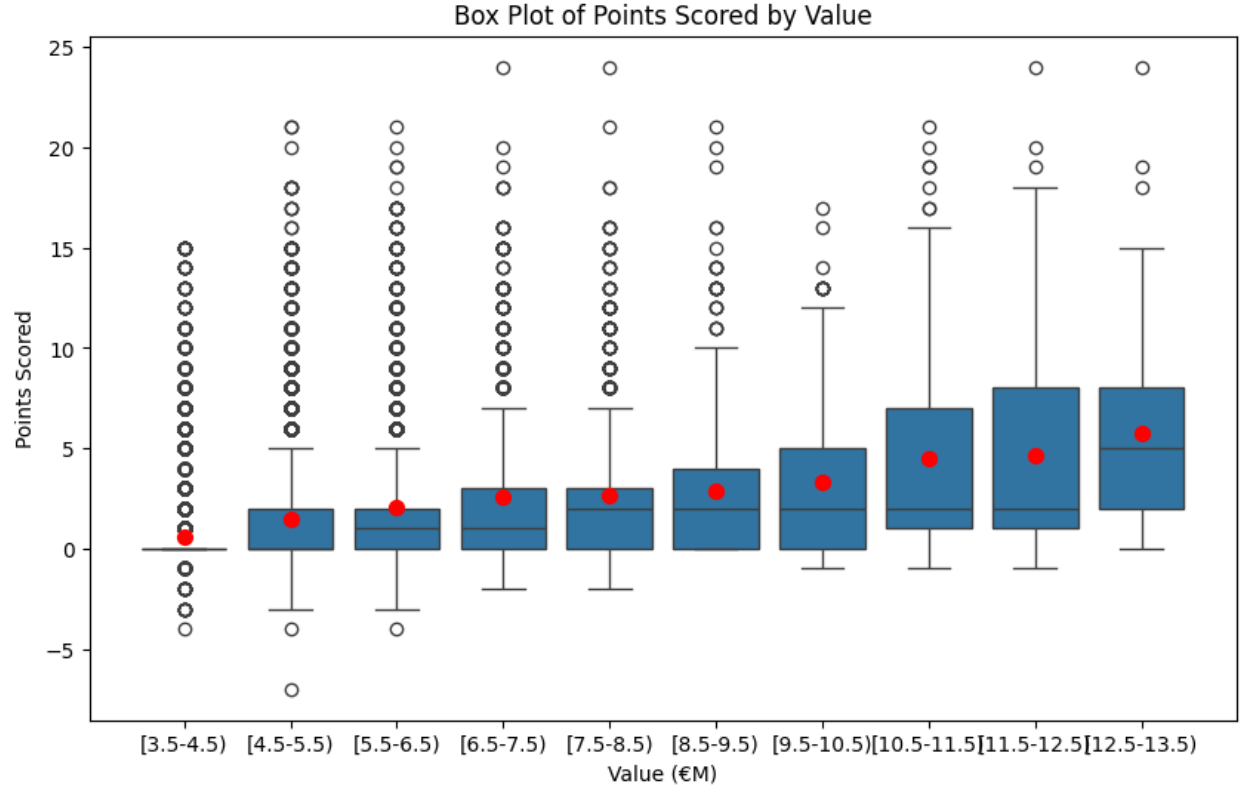


Fig 3: Points distribution with player value

4.2. Feature Importance

To understand the effect of player and team form on current performance, we plot the correlation matrix for the previous 4 gameweeks points as well as player performance indices. it is observed that autocorrelation to previous gameweek points is week with the maximum being the latest gameweek. A similar trend is observed with the performance indices such as ICT Index as well. Hence, we need to consider more variables to improve the explanation of variation.

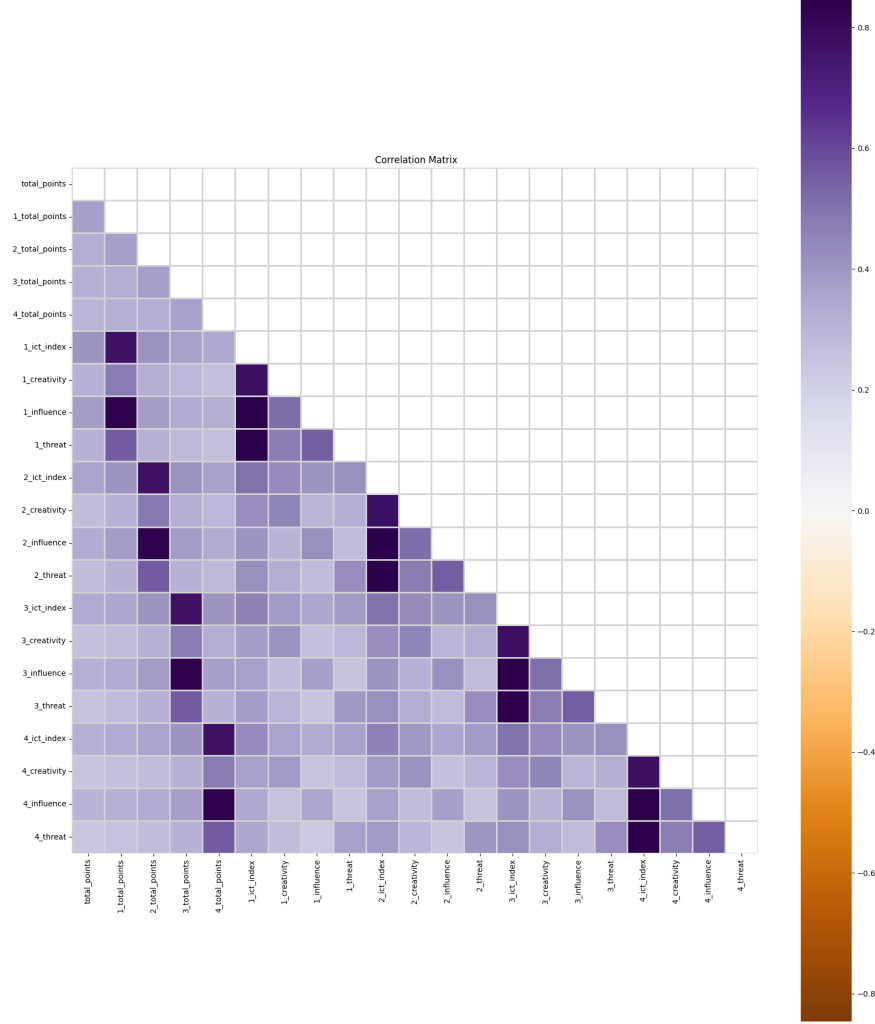


Fig 4: Correlation matrix between points and performance metrics from previous gameweeks

We also checked the impact of previous gameweek player's statistics and team statistics on total points using correlation matrix. From the two figures below, it can be understood that minutes played in previous matches have a significant contribution to the points earned because in a highly competitive league like Premier League, a player with high consistency on field only gets to play regularly and make the most out of the minutes he gets by contributing in their respective roles, such as scoring goals, providing assists, and keeping clean sheets. However, the overall team statistics is seen to have very low impact on the individual's points as football is a highly team-chemistry based game; hence, no single player can create an impact if the other players are not equally involved.

The limitation of correlation matrix given that the outcome variable total_points is right skewed, are as follows:

- Underestimating Relationship: Skewed data's extreme values can downplay true correlations, impacting relationship assessment.
- Non-linear Concerns: Correlation assumes linearity, failing to capture non-linear dependencies in skewed data.
- Outlier Sensitivity: Skewed data's outliers disproportionately affect correlation coefficients, leading to biased results.
- Missed Feature Interactions: Complex interactions in skewed data might be missed, limiting complete

feature feature understanding.

To improve gauging feature importance, a tree-based model such as Random Forest was chosen, as it can be useful for understanding variable importance, even when dealing with right-skewed data.

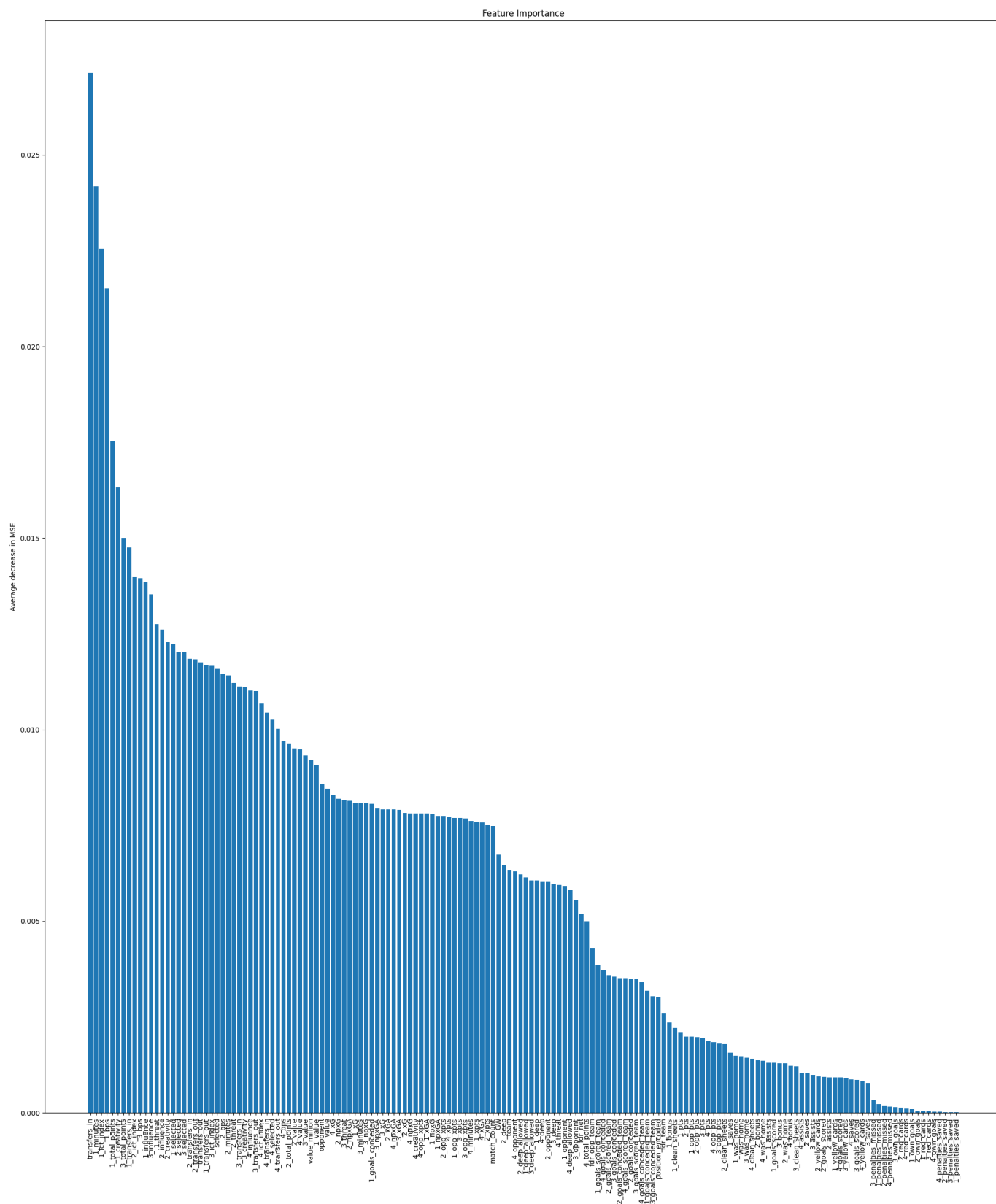


Fig 5: Feature importance from random forest regression

From the above plot, we can say that the top 10 features for predicting points are:

1. **transfers_in**: Number of transfers made to buy the player by FPL users in current gameweek.
2. **1_minutes**: Minutes played in the last gameweek.
3. **1_ict_index**: ICT index rated in the last gameweek.
4. **1_bps**: Bonus point system rating earned in the last gameweek.
5. **1_total_points**: Points earned in the last gameweek.
6. **1_creativity**: Creativity of the player in the last gameweek.
7. **3_total_points**: Points earned in the third last gameweek.
8. **1_transfers_in**: Number of transfers made to buy the player in the last gameweek.
9. **2_ict_index**: ICT Index rated in the second last game week.
10. **3_bps**: Bonus point system rating earned in the third last gameweek.

From the above feature importance output, it can be inferred that choices of FPL users on selection of players are positively correlated with the points scored by the players hence, template” teams are safe bet to go with earning slightly risk-free returns. However, further analysis needs to be done to understand the phenomenon better. Player performance metrics as suggested earlier also show up as expected.

The limitations of this approach are as follows:

- **Sensitivity to Outliers**: While Random Forests are less sensitive to outliers than linear models, extreme values in the right tail of the skewed distribution might still influence feature importance.
- **Open to Interpretation**: The feature importance plot itself does not indicate the direction of the relationship (positive or negative) between each feature and the target variable. It only indicate the magnitude of importance.

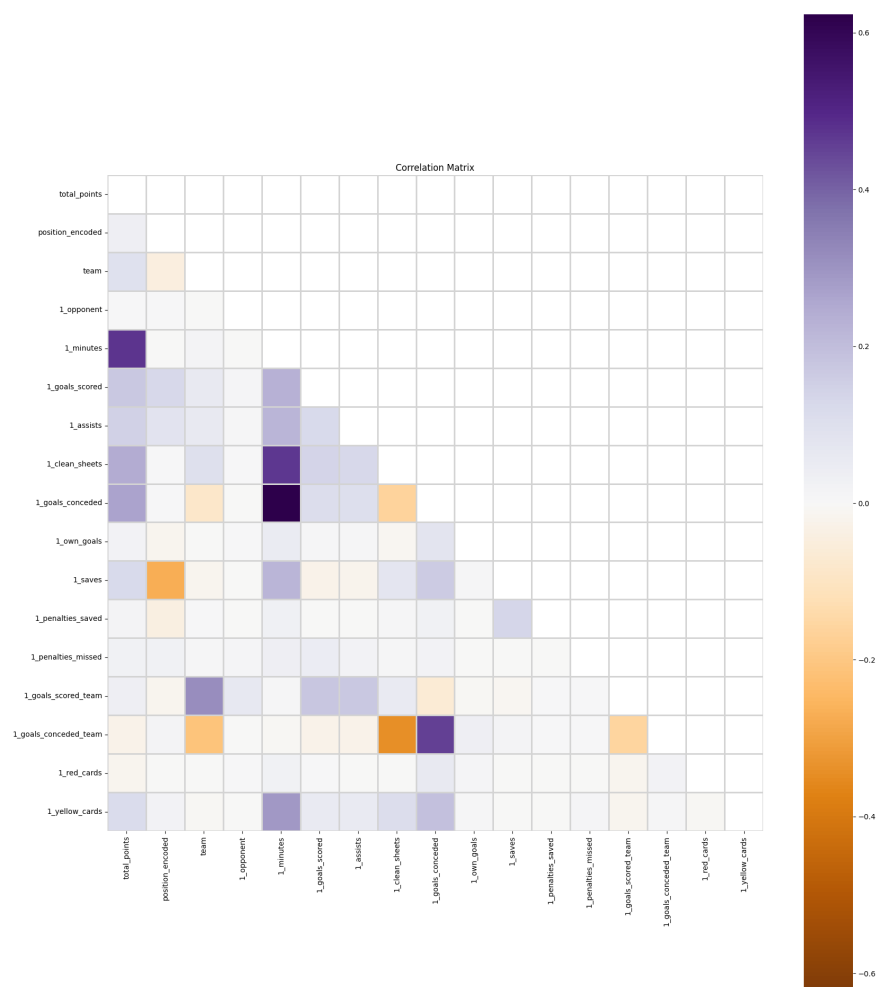


Fig 6: Correlation matrix between points and player statistics in last match

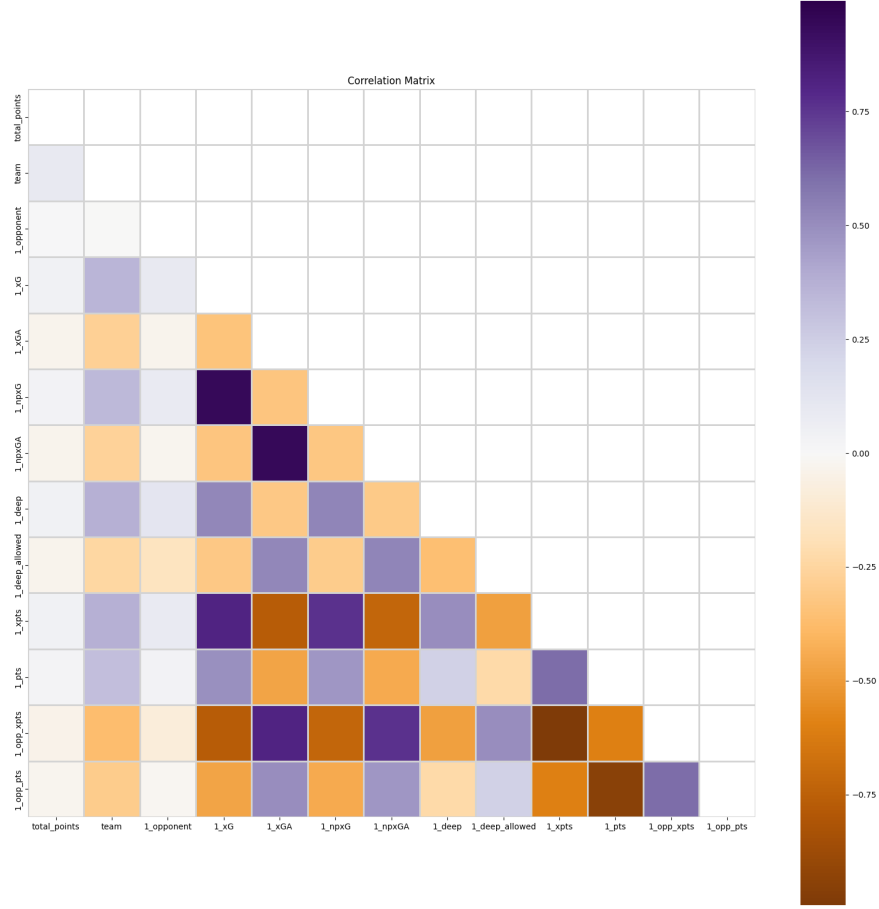


Fig 7: Correlation matrix between points and team statistics in last match

4.3. Regression model

The data set was split into two parts: the training set, which includes data from the 2020-21 and 2021-22 seasons, and the test set, which contains data from the 2022-23 season. We created this split to test whether a model trained on data from the two previous seasons can predict outcomes for the upcoming season. Within the training data set, we further created a randomly sampled split of 75% for training and 25% for validation.

4.3.1. Linear Regression

As mentioned earlier, linear relationships are not applicable because the outcome variable does not meet the assumptions of a normal distribution. However, to benchmark performance against other models, we created a ridge linear regression model for appropriate feature selection and bias balance. The final ridge regression model gave the following output:

Error	Training set	Validation set	Test set
Mean Squared Error	2.04	2.10	

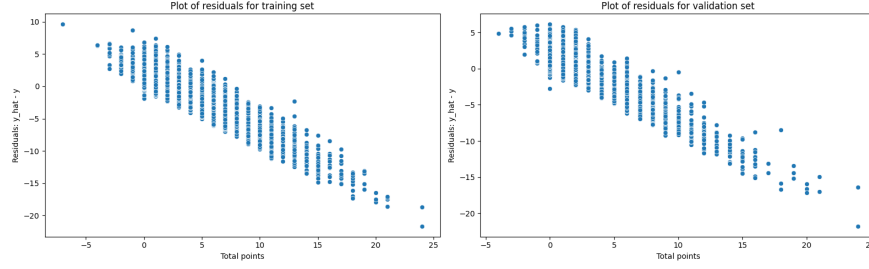


Fig 8: Residual plot for training and validation set

As suspected earlier and observed in the residual plot, we can conclude that there is significant heteroskedasticity. Therefore, a linear model may not be an appropriate choice for predicting footballer points. However, it can still serve as a benchmark for evaluating other regression models.

Given that the data is highly imbalanced, high-dimensional, noisy, and requires robustness for use in live fantasy premier league games, ensemble methods seem to be the best fit for our use case. Consequently, we have employed random forest regressor and extreme gradient boosting methods as our next steps.

4.3.2. Random Forest

An initial run with 100 estimators and maximum 5 features as parameters yielded an MSE of 2.06 for the validation set, which is already an improvement over the best trained linear model. We went ahead with parameters tuning to tweak the model for best output and obtained the following parameters:

- Number of trees in the forest `n_estimators:100`
- Number of features considered to split a node, `max_features:25`
- Maximum depth of each tree, `max_depth:40`

The final random forest regression model gave the following output:

Error	Training set	Validation set	Test set
Mean Squared Error	0.77	1.99	

It can be observed that there is some degree of overfitting in the train set when the (MSE) is compared to the validation dataset

4.3.3. Extreme Gradient Boosting(XGBoost)

Similar to random forest, we did an initial run for XGBoost regression and found that the MSE for the validation set was which surpassed both linear and random forest model.

4.4. Clustering for New Players

Addressing the issue of new players being added to the league every year is the most important challenge we aimed to solve. While there are various approaches, such as Bayesian regression using global football data, we chose clustering as the most suitable method given the limited data available on each player. Our current dataset only contains information on the player's name, position, team, new value, and details of their first match, such as the opponent, team rating, and whether it was a home or away game.

Hence, we formulated our approach in the following manner:

- Pre-process data to create dataset similar to data as available for new player, i.e a player who has played less than 5 matches in the league.

- Use clustering technique to group existing players based on attributes similar to ones available for new players such as position, value, team rating, matches played, etc.
- Once the clusters are formed, assign the new player to the cluster with most similar attributes and estimate the new player's points based on the average points of players within that cluster.

4.4.1. Data Pre-processing

For this model, the data set was refreshed to include all players from the training data set, however the data set was limited to either first four games played by the players in the league as a whole or first four games of each seasons.

4.5. Optimization

Using the expected points scored by the individual players, we aimed to select the best possible team for the future gameweek that would provide us an advantage over other fantasy players. We have implemented three approaches to test which provides the most consistent returns. A general simplification has been done all three approaches i.e to select 15 player squad and then a 11 player team which would then play during the gameweek and earn points, the problem turns out multi integer programming optimization. However, given the high complexity and computation requirement, we have simplified the problem to a two-stage integer programming optimization.

4.5.1. Maximizing Expected Points

Optimization of team selection was conducted in two stages, with the objective of maximizing the total expected points of the selection at each stage. The 1st stage involved creating a selection of a 15-member squad from a pool of available players, e.g., 572 players in gameweek 1, while enforcing certain constraints. The 2nd stage of optimization was carried out on this 15-member selection group to create the final team of 11 players under the constraints of the game. The processes are described below:

1st Stage: Selecting Squad of 15 Players

Objective:

$$\max \left\{ \sum_{i=1}^N X_i xP_i \right\}$$

where X_i is a binary selection variable, i.e., 1 if player i is selected, and 0 if not, out of N players available for selection in the gameweek. xP_i is the expected points for player i for the match.

Constraints:

1. **Size Constraint:** $\sum_{i=1}^N X_i = 15$
2. **Budget Constraint:** $\sum_{i=1}^N X_i V_i = 100$ where V_i is the value of the i th player.
3. **Position Constraints:**
 - (a) Goalkeepers: $\sum_{i=1}^N X_i GK_i = 2$, where GK_i is the binary variable, 1 if position of the player i is goalkeeper (GK) and 0 otherwise.
 - (b) Defenders: $\sum_{i=1}^N X_i DEF_i = 5$, where DEF_i is the binary variable, 1 if position of the player i is defender (DEF) and 0 otherwise.
 - (c) Midfielders: $\sum_{i=1}^N X_i MID_i = 5$, where MID_i is the binary variable, 1 if position of the player i is midfielder (MID) and 0 otherwise.
 - (d) Forwards: $\sum_{i=1}^N X_i FWD_i = 3$, where FWD_i is the binary variable, 1 if position of the player i is forward (FWD) and 0 otherwise.
4. **Team Constraint:** $\sum_{i=1}^{N_j} X_i \leq 3$, where players i belonging to team j and this holds for all teams j .
5. **Binary Constraint:** All X_i are binary variables.

2nd Stage: Selecting Team of 11 Players

Objective:

$$\max \left\{ \sum_{n=1}^{15} Y_i xP_i \right\}$$

Where Y_i is a binary selection variable, i.e., 1 if player i is selected and 0 if not, out of 15 players in the squad, xP_i represents the expected points for the match for player i .

Constraints:

1. **Size Constraint:** $\sum_{i=1}^{15} Y_i = 11$
2. **Position Constraints:**
 - (a) Goalkeepers: $\sum_{i=1}^{15} Y_i \cdot GK_i = 1$, where GK_i is a binary variable, equal to 1 if position of player i is goalkeeper(GK) and 0 otherwise.
 - (b) Defenders: $3 \leq \sum_{i=1}^{15} Y_i \cdot DEF_i \leq 5$, where DEF_i is a binary variable, equal to 1 if position of player i is defender(DEF) and 0 otherwise.
 - (c) Midfielders: $2 \leq \sum_{i=1}^{15} Y_i \cdot MID_i \leq 5$, where MID_i is a binary variable, equal to 1 if position of player i is midfielder(MID) and 0 otherwise.
 - (d) Forwards: $1 \leq \sum_{i=1}^{15} Y_i \cdot FWD_i \leq 3$, where FWD_i is a binary variable, equal to 1 if position of player i is forward(FWD) and 0 otherwise.
3. **Binary Constraint:** All Y_i are binary variables.

4.5.2. Maximizing Expected Points and Minimizing the Variance of the Points

Similar to portfolio management, optimal team selection involves not only maximizing expected points but also minimizing the risk associated with those points. The reasoning is that if a player is expected to perform well and earn high xP points, there is also a risk associated with their performance. We will measure this risk in terms of variance in the xP points scored from previous season. Accordingly a variance covariance matrix is constructed with xP points from previous season. Its noteworthy here that this matrix also takes into count the "chemistry" between team players, putting simply how well the players will perform as a team. This is taken into care by covariance terms in the non-diagonal elements in the matrix. Thus in this optimization problem, we will define the objective function as below:

Objective:

$$\max \left\{ \frac{\sum_{n=1}^N Y_i xP_i}{\sum_{i=1}^N \sum_{j=1}^N cov(xP_i, xP_j)^{\frac{1}{2}}} \right\}$$

Where Y_i is a binary selection variable i.e, 1 if player i is selected, 0 if not

5. Result and discussions

6. Conclusion

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